Sequential Circuits - Single-Input Flip-Flops (T & D)

While SR and JK flip-flops offer multiple control inputs, we often only need a circuit to do one of two specific things: act as a pure memory delay (D) or act as a frequency toggler (T). We can create these single-input flip-flops by mathematically modifying our existing dual-input circuits.

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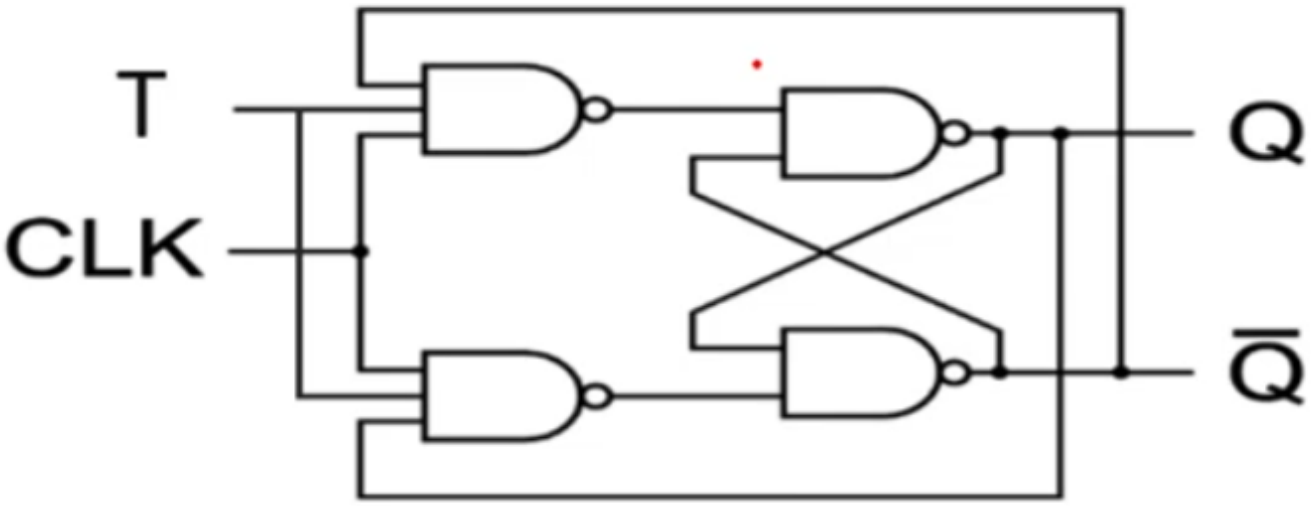
1. The T (Toggle) Flip-Flop

The T flip-flop is created by tying the J and K inputs of a JK Flip-Flop together into a single wire called T.

- If $T = 0$, both J and K receive `0`, meaning the circuit Holds its state.
- If $T = 1$, both J and K receive `1`, meaning the circuit Toggles its state on the clock edge.

Because holding $T = 1$ causes the output to invert on every single clock cycle, the T Flip-Flop is widely used to build binary counters and frequency dividers (it cuts the clock frequency exactly in half).

Circuit Diagram



Fundamental Tables

Truth Table (Function Table):

Clk	T	Q_{n+1}
0	0	Q_n (No Change)
1	1	$\overline{Q}_n$ (Toggle)

Characteristic Table:

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table:

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic Equation

If you look closely at the Characteristic Table, the output Q_{n+1} is `1` only when T and Q_n are *different*. This is the exact definition of an Exclusive-OR (XOR) gate.

$$Q_{n+1} = T \oplus Q_n$$

(Expanded: $Q*{n+1} = T\overline{Q}*n + \overline{T}Q_n$)

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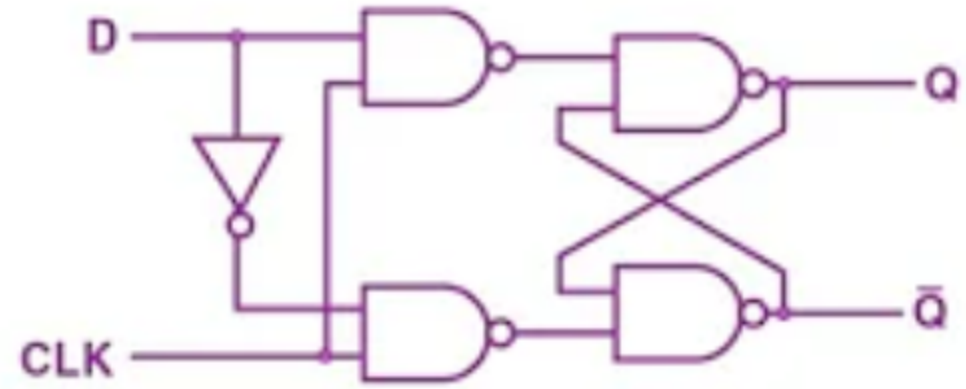
2. The D (Data/Delay) Flip-Flop

The D flip-flop is created by attaching an inverter (NOT gate) between the S and R inputs of an SR Flip-Flop, creating a single input D.

- This guarantees that S and R are always opposites, completely eliminating the Invalid $(1, 1)$ state.
- Whatever data is on the D pin is simply passed through to the Q output on the active clock edge.

Because it just captures and holds the input, it is the standard building block for computer memory (RAM) and data registers.

Circuit Diagram



Fundamental Tables

Truth Table (Function Table):

Clk	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

Characteristic Table:

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table:

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Characteristic Equation

Because the next state (Q_{n+1}) completely ignores the current state (Q_n) and simply copies the D input exactly, the equation is incredibly simple:

$$Q_{n+1} = D$$